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PREDICTING GREEN SPACE OF A LOCATION THROUGH
HISTORICAL IMAGE ANALYSIS AND PREDICTIVE MODELING

B.Sc. ENGINEERING THESIS

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DECLARATION

We hereby declare that the study reported in this thesis entitled as above is our own original work and has not been submitted before anywhere for any degree or other purposes. Further, we certify that the intellectual content of this thesis is the product of our own work and that all the assistance received in preparing this thesis and sources have been acknowledged and cited in the reference Section.

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ABSTRACT

Title in Bold with Center Alignment

In the race of urbanization, the cities have lost their greenery. Cities are becoming densely populated with time and green spaces are shrinking fast. There is a growing need for various services, such as housing, workplaces and other facilities such as schools and hospitals, to meet the needs of the local population. The forests are also impacted; the damage is not limited to metropolitan areas. Forest trees are being chopped down for a variety of reasons, including the production of wood and other building materials, fuel, land development, and plenty more. Urban planners put a lot of effort into figuring out how to use every acre of land productively when designing urban landscapes. Environmental factors are therefore disregarded. However, the negligence has reached a point where there is no choice but to restore the damage and integrate green space into communities. Hence, the objective of this research is to propose a predictive system which can give a predictive value of the green view index of an area based on its history of images of the past years at 5-year intervals. To achieve this objective, firstly, traditional deep learning and ensemble Learning techniques were implemented with a publicly available and self-made data sets containing 5108 and 1705 images respectively. All the models were evaluated using suitable performance metrics and the ensemble learning approach was found to perform better (**with accuracy, precision, and dice-coefficient of 0.94, 0.96 and 0.91**). Among the four RNN models trained, LSTM achieved the lowest error metrics (**MAE of 0.2707, MSE of 0.1492 and an RMSE of 0.3862**). Finally, a two-module framework is proposed to forecast the change in a location's future green view index consisting of individual Siamese Models and an LSTM Model. By leveraging satellite imagery and time series data for green space detection and prediction, a contribution can be made to offer insights into urban planning and environmental conservation.

সারসংক্ষেপ

Predicting Green Space of a Location Through Historical Image Analysis And Predictive Modeling

নগরায়নের দৌড়ে শহরগুলো হারিয়েছে তাদের সবুজ। সময়ের সাথে সাথে শহরগুলি ঘনবসতিপূর্ণ হয়ে উঠছে এবং সবুজ স্থানগুলি দ্রুত সঙ্কুচিত হচ্ছে। স্থানীয় জনসংখ্যার চাহিদা মেটাতে বিভিন্ন পরিষেবা, যেমন আবাসন, কর্মক্ষেত্র এবং অন্যান্য সুযোগ-সুবিধা যেমন স্কুল ও হাসপাতালের চাহিদা বাড়ছে। বনও প্রভাবিত হচ্ছে; ক্ষতি শুধু মেট্রোপলিটন এলাকায় সীমাবদ্ধ নয়। কাঠ ও অন্যান্য নির্মাণ সামগ্রী উৎপাদন, জ্বালানি, ভূমি উন্নয়ন এবং আরও অনেক কিছু সহ বিভিন্ন কারণে বনের গাছ কেটে ফেলা হচ্ছে। নগর পরিকল্পনাবিদরা শহরে ল্যান্ডস্কেপ ডিজাইন করার সময় কীভাবে প্রতিটি একর জমি উৎপাদনশীলভাবে ব্যবহার করা যায় তা খুঁজে বের করার জন্য অনেক প্রচেষ্টা করছেন। তাই পরিবেশগত কারণ উপেক্ষিত হচ্ছে। যাইহোক, অবহেলা এমন এক পর্যায়ে পৌঁছেছে যেখানে ক্ষতি পুনরুদ্ধার করা এবং সম্প্রদায়ের মধ্যে সবুজ স্থানকে একীভূত করা ছাড়া কোন উপায় নেই। তাই, এই গবেষণার উদ্দেশ্য হল একটি ভবিষ্যদ্বাণীমূলক ব্যবস্থার প্রস্তাব করা যা 5 বছরের ব্যবধানে বিগত বছরের ছবিগুলির ইতিহাসের উপর ভিত্তি করে একটি এলাকার সবুজ দৃশ্য সূচকের একটি ভবিষ্যদ্বাণীমূলক মান দিতে পারে। এই উদ্দেশ্য অর্জনের জন্য, প্রথমত, ঐতিহ্যগতভাবে Deep Learning এবং Ensemble Learning কৌশলগুলি Publicly available এবং Self-made যথাক্রমে 5108 এবং 1705 ছবি সম্বলিত ডেটা সেট তৈরি করেছে। সব মডেলগুলি উপযুক্ত কর্মক্ষমতা মেট্রিক্স ব্যবহার করে মূল্যায়ন করা হয়েছে এবং Ensemble Learning পদ্ধতিটি সম্পাদন করার জন্য পাওয়া গেছে (with accuracy, precision, and dice-coefficient of 0.94, 0.96 and 0.91)। প্রশিক্ষিত চারটি RNN মডেলের মধ্যে, LSTM সর্বনিম্ন ত্রুটির মেট্রিক্স 0.2707 এর MAE, 0.1492 এর MSE এবং 0.3862 এর RMSE অর্জন করেছে। অবশেষে, পৃথক Siamese Models এবং একটি LSTM মডেল সমন্বিত একটি অবস্থানের ভবিষ্যতের সবুজ দৃশ্য সূচকে পরিবর্তনের পূর্বাভাস দেওয়ার জন্য একটি দ্বি-মডিউল কাঠামোর প্রস্তাব করা হয়েছে। স্যাটেলাইট ইমেজ এবং সবুজ স্থান সনাক্তকরণ এবং ভবিষ্যদ্বাণীর জন্য টাইম সিরিজ মডেল তৈরি করা হয়েছে, যা একটি অবদান হতে পারে নগর পরিকল্পনা এবং পরিবেশ সংরক্ষণ বিষয়ে অন্তর্দৃষ্টি প্রদানের জন্য।

LIST OF MAIN NOTATION

<i>CNN</i>	Convolutional Neural Network
<i>GVI</i>	Green View Index
<i>GSV</i>	Google Street View
<i>ML</i>	Machine Learning
<i>DCNN</i>	Deep Convolutional Neural Networks
<i>UCGS</i>	Urban Community Green Space
<i>HRNet</i>	High-Resolution Net
<i>ASPP</i>	Atrous Spatial Pyramid Pooling
<i>LSTM</i>	Long Short-Term Memory
<i>ATT – LSTM</i>	Attention Long Short-Term Memory
<i>GRU</i>	Gated Recurrent Unit
<i>ATT – GRU</i>	Attention Gated Recurrent Unit
<i>RMSE</i>	Root Mean Square Error
<i>RNN</i>	Recurrent Neural Network
<i>MAE</i>	Mean Absolute Error
<i>MSE</i>	Mean Square Error

slinetocsectionLIST OF TABLES

LIST OF TABLES

LIST OF FIGURES

TABLE OF CONTENTS

CHAPTER 1

INTRODUCTION

This chapter discusses

1.1 Research Background

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1.2 Thesis Objectives

The following are the objectives of this thesis:

1. To develop a model that will detect green space from aerial images.
2. To propose a framework architecture for predicting the green space of a location in the next five years.
3. To perform a comparative analysis of the outcomes of the trained models.

1.3 Methodological Overview

To achieve the above objectives, this research was conducted in two phases - segmentation and prediction. Firstly, a literature study was conducted to understand how green space is being measured, the approaches followed to segment green spaces from images, what studies have been conducted so far related to forecasting green spaces in a designated area, and the methodology framework that has been defined. Secondly, the models

and techniques followed for the segmentation and prediction of green space have been proposed. Finally, a comparative analysis was conducted to evaluate the performance of the conducted approaches and the best-performed system is proposed for predicting green spaces of a location from historical aerial images.

1.4 Scope of The Thesis

The thesis primarily focused on developing a system that can predict the green space of a given location based on the historical aerial images of 15 years. The models were trained on aerial images. Therefore, the system is not designed for ground-level view. Moreover, images with a gap of 5 years were used for training the model. So, the system requires images with a gap of 5 years to give appropriate results. Additionally, the system is designed to give a numerical value (GVI difference value) as output.

1.5 Organization of The Thesis

The remainder of this research is organized as follows. The theoretical background and related works are briefly introduced in Chapter 2.

The methodology which was followed throughout this research is presented in Chapter 3. The overall work has been divided into two phases (segmentation and prediction). Methodology of both the phases and the final proposed framework are described in brief.

The overall evaluation of the system is discussed in Chapter 4. The evaluation consists of result analysis and comparative analysis of the functionalities and representation of both phases of the proposed framework.

The engineering considerations, challenges and remedies involved in this thesis have been discussed in Chapter 5.

Finally, the discussion and conclusion are discussed in chapter 6. The discussion includes the thesis overview, contribution, research outcome, limitations, and future works.

CHAPTER 2

THEORETICAL BACKGROUND AND LITERATURE

REVIEW

This chapter explores previous works and approaches. It also discusses how modern techniques like image processing and machine learning are used for developing quick and smart identification and forecasting technologies. This chapter comes to an end with a critical appraisal that highlights the research gaps.

2.1 Green View Index

One of the indices available for quantifying roadside greening is the Green View Index (GVI). It was first proposed by ? (?). The green view index (GVI) uses an observer's visual perception to quantify the urban space by simulating the pedestrian perspective of the road in street-view photo data and then calculating the percentage of vegetation in the road landscape. It is an effective way to acquire what people actually and typically see at street level on the ground. It is defined as the ratio of the total green area from four pictures taken at a street intersection to the total area of the four pictures, calculated using the following equation:

$$\text{Green View} = \frac{\sum_{i=1}^4 \text{Area}_{g_i}}{\sum_{i=1}^4 \text{Area}_{t_i}} \times 100\% \quad (2.1)$$

where Area_{g_i} is the number of green pixels in the picture taken in the i th direction among the four directions (north, east, south and west) for one intersection, and Area_{t_i} is the total pixel number of the picture taken in the i th direction.

? (?) modified the GVI and used six images covering the 360° horizontal surroundings to calculate the index for each sample site along streets. The final modified GVI used calculated using 18 images for each site. The modified GVI calculation formula is written as

$$\text{Green View} = \frac{\sum_{i=1}^6 \sum_{j=1}^3 \text{Area}_{g_{ij}}}{\sum_{i=1}^6 \sum_{j=1}^3 \text{Area}_{t_{ij}}} \times 100\% \quad (2.2)$$

where $\text{Area}_{g_{ij}}$ is the number of green pixels in one of these images captured in six directions with three vertical view angles (45°, 0°, 45°) for each sample site, and $\text{Area}_{t_{ij}}$ is the total pixel number in one of the 18 images.

2.2 Aerial Image

An aerial image specifically refers to a photograph or digital image captured from an elevated perspective, typically obtained via drones or satellites. These images serve as crucial data inputs for analyzing and mapping various features, including land cover, vegetation, and environmental changes, facilitating the application of deep learning techniques for tasks such as green space identification and segmentation. Figure ?? shows how aerial images differ from street-view images.

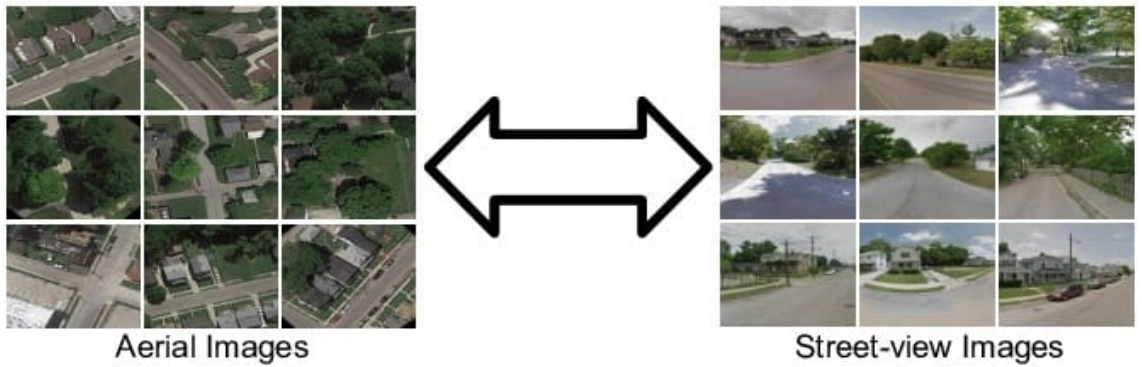


Figure 2.1: Aerial Image vs Street-View Image

2.3 Digital Image Processing

Digital Image Processing (DIP) is the manipulation and analysis of images using computer algorithms. It involves enhancing, analyzing, and extracting information from digital images to improve their quality or extract meaningful data. Techniques like image enhancement, image restoration, object detection, image segmentation, and pattern recognition fall under DIP.

2.4 Semantic Segmentation

Semantic segmentation is a type of deep learning system that gives each pixel in an image a label or category. It is used to distinguish between a set of pixels that belong to distinct categories (e.g., sky, water, person). For example, an autonomous vehicle needs to be able to identify other vehicles, people, pavement, traffic signs, and other features of the path. Semantic segmentation is used in many applications such as automated driving, medical imaging, industrial inspection and object detection, etc.

A simple example of semantic segmentation is separating the images into two classes. For example, in Figure 2.2, an image showing a person at the beach is paired with a version showing the image's pixels segmented into two separate classes: person and background.



Figure 2.2: Semantic Segmentation

2.5 Machine Learning

The goal of machine learning (ML) is to comprehend and develop "learning" techniques, or tactics that use data to improve performance on specific tasks. (1, 2). That's regarded as a part of artificial intelligence. Machine learning algorithms use sample data, also known as training data, to build a model using which they may make predictions or judgments without having to be taught (3, 4). In many fields, such as computer vision, speech recognition, email filtering, health, and agriculture, where it is difficult or impractical to develop conventional algorithms that can complete the necessary tasks, machine learning algorithms are employed. (5, 6, 7). Performing machine learning can involve creating a model, which is trained on some training data and then can process additional data to make predictions. The machine learning models used in this thesis fall under Deep Learning, which is a new field of study in machine learning. It is made up of several artificial neural network hidden layers. (8, 9)

2.6 Deep Learning Architectures

Deep Learning models are based on Artificial Neural Networks. These neural networks are inspired by the structure and function of the human brain's biological neurons and are modeled to learn from a vast number of data. Due to the availability of enormous datasets and advancements in computing power, the popularity of Deep Learning has grown in recent years (10, 11) The deep learning architectures used in this thesis are described below.

2.6.1 U-Net

U-Net is a popular architecture in satellite image analysis. As shown in Figure ?? U-Net's architecture has two paths: a contracting road and an expanding method (12, 13).

The expanding path contains decoder layers that decode the encoded data and use the information from the contracting path via skip connections to generate a segmentation map. The contracting path lowers the spatial resolution of the input and finds the relevant features in the supplied image. The encoder layers perform convolutional operations that reduce the spatial resolution of the feature maps while increasing their depth to capture increasingly abstract representations of the input.

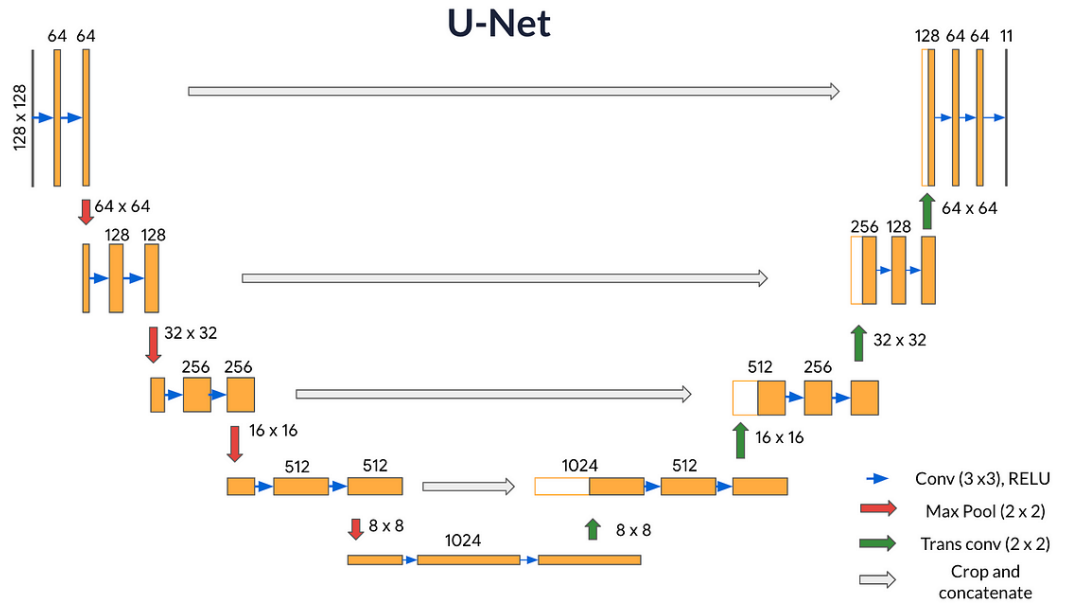


Figure 2.3: U-Net

2.6.2 SegNet

This novel and practical deep fully convolutional neural network architecture for semantic pixel-wise segmentation was proposed by ? (?). As shown in Figure ??, Segnet uses an encoder network and a corresponding decoder network comes before a pixel-wise classification layer. SegNet only requires forward evaluation of a fully learned function to obtain smooth label predictions. Additionally, a larger context is considered for pixel labeling which improves accuracy. Moreover, it is easy to visualize the effect of feature activation(s) in the pixel label space at any depth.(?, ?)

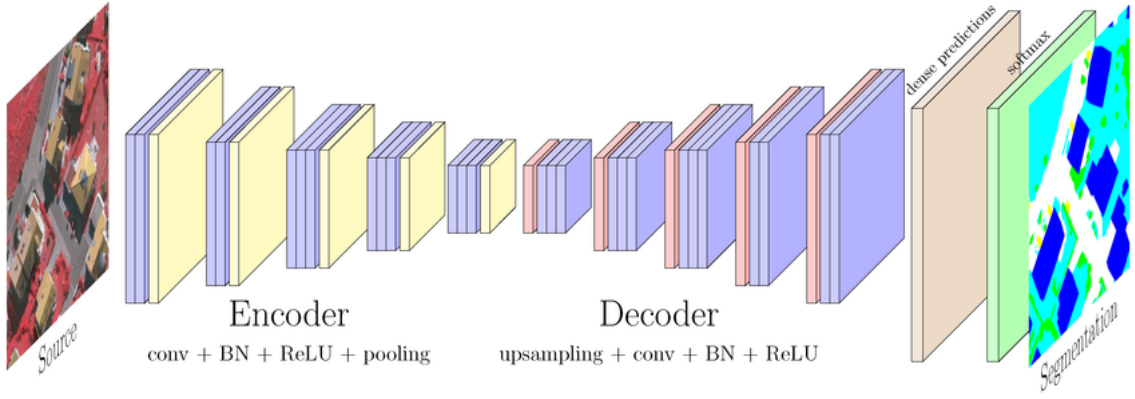


Figure 2.4: Segnet Architecture

2.6.3 DeepLabv3plus

DeepLabV3plus is an advanced deep neural network architecture for semantic image segmentation which improves upon DeepLabV3 by integrating multi-scale feature fusion and Atrous Spatial Pyramid Pooling (ASPP) modules for better context understanding (?, ?). Atrous Spatial Pyramid Pooling efficiently harvests multi-scale characteristics with valuable information for segmentation (?, ?). As shown in Figure ??, the input image is encoded into a compressed representation using an encoder, and then these characteristics are upsampled to the required resolution using a decoder. With its encoder-decoder architecture and skip connections, it achieves high accuracy in segmenting objects of varying sizes.

2.7 Recurrent Neural Network

RNN, or Recurrent Neural Network, is a class of neural networks specially designed for processing sequential data by maintaining an internal state or memory. Unlike feedforward neural networks, which process data in a single direction, RNNs have connections that loop back, allowing them to exhibit temporal dynamics and learn patterns over time.

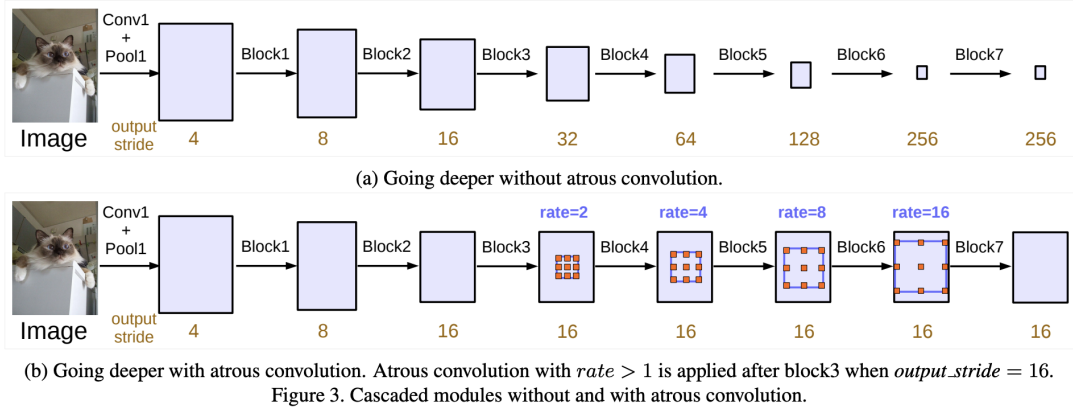


Figure 2.5: Deeplabv3 Architecture

RNNs are widely used in tasks such as language modeling, speech recognition, and time series prediction. RNN has variants like LSTM and GRU.

2.7.1 LSTM (Long Short-Term Memory)

Long Short-Term Memory is a type of computer algorithm used for processing sequences of data, like sentences or time-series data. It's special because it can remember important information for a long time while also quickly forgetting irrelevant details. This makes it great for understanding and predicting patterns in sequences, such as in natural language processing tasks like language translation or sentiment analysis (?, ?)

2.7.2 GRU (Gated Recurrent Unit)

Gated Recurrent Unit, is a type of recurrent neural network (RNN) architecture designed to address the vanishing gradient problem encountered in traditional RNNs. GRU incorporates gating mechanisms to regulate the flow of information through the network, allowing it to retain long-term dependencies more effectively. With fewer parameters compared to other RNN variants like LSTM (Long Short-Term Memory), GRU offers computational efficiency while achieving comparable performance in sequence modeling

tasks such as natural language processing and time series prediction. (?, ?)

2.7.3 Siamese Neural Network Model

A Siamese model is a neural network architecture designed for comparing similarity between pairs of inputs. It consists of two identical subnetworks (or twins) that share the same weights and architecture. Each input is processed independently by one of the subnetworks, and the outputs are then compared using a distance metric. Siamese Neural Networks are a type of neural network design consisting of two or more identical subnetworks. In this case, "identical" means that their weights and parameters are the same. The parameter updates are mimicked by the two sub-networks. Because these networks can compare feature vectors to ascertain how similar the inputs are, they are helpful in a wide range of applications (?, ?). Siamese models are commonly used in tasks like image recognition, signature verification, and similarity-based recommendation systems.

2.8 Related Works

The existing research works have been divided into four sections: measuring green space, segmenting green space from images, analyzing change in land and predicting change in land coverage. There are two categories for segmentation works: segmentation with Google Street View photos and segmentation using aerial images. This section has studied, analyzed, and discussed existing works.

2.8.1 Measuring Green Space from Images

? (?) proposed a new index called Green View (GV) in their paper that measures the amount of greenery that people can see on the ground at different locations in a city. The researchers tested the Green View index in Berkeley, California. Their research combines

field surveys and photography interpretation. The image was first segmented into image objects using a multi-resolution segmentation algorithm. After segmentation, an iterative classification process was run to classify image objects into land cover classes until the classification results could not be improved by more iteration. Six land cover classes were identified and the accuracy of the classification result was determined by using a reference data set which are patches of different land cover types identified from the image and verified on the ground. Green View was measured using a two-step process. Sampling sites were selected based on a random sampling approach using a street map. Secondly, pictures were taken at selected intersections in four directions. Green vegetation areas in pictures were quantified by counting green pixels. The green view index was computed by manually extracting only the green pixels using Adobe Photoshop for the image and then the ratio of green pixels over total pixels was calculated. Green View values were calculated for each intersection and averaged for each census tract. Overall Green View for Berkeley was determined by weighting values based on the number of intersections in each tract. The authors successfully concluded that the overall Green View value of their subject location was 24.79%, indicating the visibility of urban forests.

Based on the previous research, ? (?) modified the existing Green View Index (GVI) formula and conducted a case study assessment of street greenery using GSV images in the area of East Village, Manhattan District, New York City. They explored Google Street View (GSV) as a street-level, urban greenery assessment tool. They used 18 GSV images from different angles to create a more objective measurement. This covers a 360° horizontal view and 180° vertical profile, offering a comprehensive assessment of street greenery. From the images, green vegetation was extracted using the difference of Green band from Red and Blue band. An additional rule that pixel values in green band must be larger than those in the red band. The green vegetation extraction results were then compared with manually extracted green vegetation for the same image using Adobe Photoshop.

In another study, ? (?) also utilized Google Street View (GSV) and remote sensing data

to estimate street tree shade provision in Boston during the leaf-on season. Using the sky view factor (SVF) to quantify shade, the research mapped the spatial distribution and results found that street trees contribute 24.6% of shade citywide, with the southwestern part exhibiting the highest coverage. From their research, the authors concluded that GSV images are qualified for assessing street greenery and the modified GVI may be a more objective measurement of street-level greenery. However, using GSV images has some limitations that include time inconsistency and coverage of a limited number of viewing points.

Similarly, ? (?) acquired a hyperspectral dataset of Brussels and created a spectral library which was used to classify the urban green types in the hyperspectral imagery through a spectral angle mapper algorithm. The study successfully identified and classified various types of urban green areas in Brussels with high accuracy.

? (?) explored the use of Google Street View (GSV) for virtual street tree surveys in municipal forestry. The research indicated GSV's potential for reliable genus identification and compared GSV data with field survey results, emphasizing GSV's promise as a geospatial tool for urban forestry research.

? (?) also used Google Street View images to analyze and quantify green canopy coverage along Singapore's road network. Results showed that trees shaded 13% of annual solar radiation, with notable variations across urban land-use types, and showcased a correlation between human and machine estimates of canopy cover.

In another similar work, ? (?) assessed street greenery using three indicators: Green View Index (GVI), Normalized Difference Vegetation Index (NDVI), and newly introduced Vegetation Structural Diversity (VSD). To gather data on street greenery in the urban area of Nanjing, China they have utilized street view images and satellite images. A semantic segmentation method was applied to extract GVI and VSD from street view images, while NDVI was calculated using satellite images. It also compared overhead greenery and eye-level greenery based on different Urban Functional Zones (UFZs) such

as residential, open spaces and industrial lands. They also made a comparison on how the indicators perform in new city vs old city. The findings from their research suggests that indicators of street greenery can provide a comprehensive reference for building more human-friendly and diversified street green belt.

2.8.2 Segmentation using Machine Learning approach

2.8.2.1 Using Google Street View image

? (?) applied state-of-the-art deep learning models, and compared their performance to a previously established benchmark of an unsupervised method. They also trained a DCNN end-to-end model that directly estimates GVI in GSV images. To validate accuracy in estimating GVI, the authors used 100 labeled GSV images and calculated Pearson's correlation coefficient between GVI from labeled images and model estimates. By using street-level GSV images and DCNN models, researchers can utilize vast street-level image datasets to learn and understand complex environmental and socio-economic phenomena, in addition to urban tree cover.

Similarly, ? (?) used Google Street View images of New York and Boston to segment and quantify the percentage of tree cover in streetscape images using a multi-step computer vision algorithm. By modeling the relationship between neighboring images along city street segments, the authors successfully estimated the amount of perceived tree cover in the city streetscapes to a relatively high level of accuracy.

2.8.2.2 Using Satellite Images

? (?) created a neighborhood-scale urban community green space (UCGS) dataset and proposed a segmentation decoder with two auxiliary branches for HRNet. Their proposed model added two additional branches to the low-resolution representations to improve

their discriminative ability, thus enhancing the overall performance when the high- and low-resolution representations are fused. To evaluate the performance of the model, it was tested on a dataset that includes satellite images of Shanghai, China. Pixel-based categorization approaches such as Maximum Likelihood (ML) and Random Forest (RF) were also used to compare the proposed model's performance. The precision, recall, F1-score, IoU, and Overall Accuracy of all the approaches were compared. The proposed model showed the most accurate UCGS detection accuracy among all comparison models. In particular, it achieved a great improvement in the identification of green spaces covered by shadows.

On the other hand, ? (?) addressed three different tasks which included land classification from satellite images using semantic segmentation. For the tasks, they created a dataset of 10,000 satellite images. They performed pixel-wise classification on their data by designing a CNN architecture based on DeepLab using ResNet18 backbone with atrous spatial pyramid pooling (ASPP) block and batch normalization. Data augmentation was performed by integrating rotations and also weighted classes based on class distributions. This approach achieved an IoU score of 0.433. Their work contains some challenges in classification including distinguishing between small structures not annotated in the ground truth and handling harder distinctions between land cover classes.

2.8.3 Analysis of Change in Land

? (?) developed a deep mapping approach using a Siamese Convolutional Neural Network (SCNN) to detect visual changes indicative of gentrification in urban areas, focusing specifically on individual properties in Ottawa, Canada. They utilized temporal sequences of Google Street View (GSV) images to track improvements in frontage quality over time. By employing the SCNN, they achieved a high test accuracy of 95.6% in detecting gentrification-like visual changes. The researchers were able to identify the

precise places and periods when gentrification processes were taking place in the study area. Insights into the spatial distribution of gentrification processes were obtained by the mapping of spatial concentrations of visual property improvements made easier by the application of Kernel Density Estimation (KDE).

In the same way, ? (?) conducted a study on gentrification in the Canadian city of Toronto, exploring its coevolution in both residential and non-residential sectors. They combined multiple data sources to analyze gentrification patterns, including household income, educational attainment, building permits, and establishment compositions. To characterize spatial units based on a composite measure of the gentrification process they employed mas-p-regions clustering. A Bayesian hierarchical spatial (BHS) model was developed to describe changes in residential property prices over five years. Their research offered an in-depth discussion of the causes and defining characteristics of gentrification, as well as insights into the coevolution of the phenomenon in both the residential and non-residential sectors.

? (?) analyzed annual forest loss data from the University of Maryland and Global Forest Watch. They used different confidence levels to distinguish between forest loss which is caused directly by fire and other causes like deforestation or natural events. A baseline of areas was established with $> 30\%$ tree coverage density in 2000 and primary forest loss was identified. They successfully identified deforestation hotspots by conducting a kernel density estimate analysis, which calculates the magnitude per unit area of the forest cover loss, using different parameters and tools.

In another work, ? (?) utilized Convolutional Neural Networks (CNNs) to analyze high-resolution RGB imagery captured by Unmanned Aerial Vehicles (UAVs) for mapping vegetation cover and species distribution. A straightforward CNN architecture was employed. Mapping tree species cover in primary forests, identifying plant invasions by woody species, and mapping vegetation succession in a glacier foreland are the main fo-

cus of their research. They achieved commendable accuracies across all case studies, with the accuracy increasing with larger tile sizes, capturing a broader spatial context. Moreover, they demonstrated the efficacy of CNNs in harnessing high-resolution data from UAVs to map vegetation patterns accurately.

? (?) have utilized UAV technology and convolution Neural Networks (CNNs) to map vegetation patterns in Dhaka city. The research formulated to calculate changes in Green spaces in Dhaka City between 1989 and 2020 using GIS and Remote Sensing techniques. Using remote sensing techniques, classification and NDVI methods of satellite imagery, they have found a drastic reduction in green spaces over the past three decades, particularly healthy vegetation, alongside urban infrastructure expansion.

2.8.4 Predicting Change in Land Coverage

? (?) used a hybrid model called the Multi-Layer Perceptron-Markov Chain (MLP-MC) to study changes in land use and land cover (LULC) in regions of Saudi Arabia from 1990 to 2018 and then predict future scenarios for 2040. The authors collected Landsat satellite images for 1990, 2000, and 2018, and used a Maximum Likelihood Classifier (MLC) algorithm to classify the images into different land cover classes. To evaluate the accuracy of these classified maps, a cross-classification method and field surveys were utilized. For predicting future changes, the Markov Chain analysis was applied to estimate transition matrices for different years, calculating LULC transition potential maps. The accuracy of their predictions was validated using measures like overall accuracy, kappa coefficient, Land Absorption Coefficient (LAC), and Land Consumption Rate (LCR). The overall classification accuracy for the years 1990, 2000, and 2018 ranged from 88.40% to 91.21%, with corresponding kappa coefficients indicating satisfactory accuracy.

Similarly, ? (?) studied how population growth and migration in the lower Himalayan regions of Pakistan led to urbanization, changing land use and increasing land surface

temperature (LST) from 1990 to 2017. They used Landsat data and the support vector machine (SVM) method to analyze these changes. For predicting future trends in 2032 and 2047, they developed a model called CA-ANN, combining cellular automata and artificial neural network, using data from 2002 and 2017, achieving a correctness value of 70% .

In another study, authors ? (?) studied how land cover (like forests and fields) changed in the Javadi Hills, India, between 2009 and 2015. They used satellite images from 2009, 2012, and 2015 and classified the images using Random Forest Classification to see how the land cover changed. Next, they compared two different models, the Markov chain-artificial neural network with cellular automata (MC-ANN-CA) and Markov chain-logistic regression with cellular automata (MC-LR-CA) to predict how the land cover might change in 2021 and 2027. They also considered factors like slope, elevation, and distance to roads to help with their predictions. Finally, they compared their predictions to real data from 2015 to see how accurate they were. The research found that the combination of artificial neural network and cellular automata was more accurate (correctness of 93.68% and the kappa value of 0.85).

Likewise, ? (?) aimed to assess urban green space (UGS) loss due to land-use land-cover (LULC) changes in Kolkata, focusing on past, present and future scenarios. Landsat imagery was used for LULC classification, which was then verified for accuracy. The CA-Markov model was employed to simulate LULC changes for 2025 and 2035. The Land use dynamic degree index (LUDD) and UGS change intensity index (CII) were calculated respectively along with Per capita urban green space (PCG) to analyze the availability of UGS in different wards of Kolkata. The research successfully evaluated UGS loss in Kolkata using modeling methods and remote sensing data, offering insightful information for green space management and urban planning.

In the same manner, ? (?) utilized remote sensing methods to analyze spatial and

temporal terrain changes. They used satellite images of Samsun Center and its coastal districts for the years 2000, 2010, and 2020 and employed the MLC algorithm for classification. The study identified four classes: Water, Built-up, Vegetation and Open Land. They used the artificial neural network (MLP-ANN) algorithm for predictive analysis. To evaluate the predictive model's accuracy, they simulated the LU/LC for 2020 using the transition potential matrix of 2000 and 2010 data, achieving a 72% accuracy based on the kappa fit index. Their work successfully forecasted potential future ecological challenges and the emergence of unproductive landscapes.

2.9 Critical Summary

In summary, several concerns were observed through the existing research. Firstly, while semantic segmentation has been used to detect green spaces from images, no research work was found that used an ensemble learning approach to identify green spaces. Additionally, even though studies have been conducted on the evolution of certain features of an area over time, such as changes in land or urbanization, very little research was found on predicting any of the studied features. Moreover, the frameworks followed for the predictive analysis have potential limitations as they did not utilize a time-series data analysis approach. Therefore, this study aims to use an ensemble technique approach to segment green space from aerial images and then forecast a measurable value to estimate the change in green space of a chosen location over the next five years.

CHAPTER 3

METHODOLOGY

The first and foremost task of the research was image acquisition which contains components such as image collection, preprocessing, and augmentation. The research is divided into two phases - detection and prediction. The detection was conducted in two techniques: Training Individual Models and Ensemble Learning Approach. All combinations of trained models were evaluated and then a comparative analysis of the models was assessed. Similarly, the prediction phase followed the following workflow - image pairing and calculation of GVI difference, training of the selected RNN models, and evaluation of their performance to select the best performing and finally, predicting the future GVI difference using siamese model and the best performing RNN model. Figure ?? illustrates the methodology followed for detection while Figure ?? portrays the procedure of the prediction phase. All the proceedings are elaborately discussed below.

3.1 Data Acquisition

The acquisition of required image data was the first focus of this research. Two primary data sets have been utilized to achieve this - a pre-existing data set from Kaggle (?, ?) and a custom data set. The pre-existing data set consisted of 5108 aerial forest images, while the custom data set was created from aerial images of different locations using Google Earth Pro. The custom data set consists of data from different types of locations such as forest locations, residential locations, and parks from urban locations. To ensure uniformity, all images from both the Kaggle data set and the custom data set are resized to a standardized dimension of 256x256. By resizing the images to an identical size,

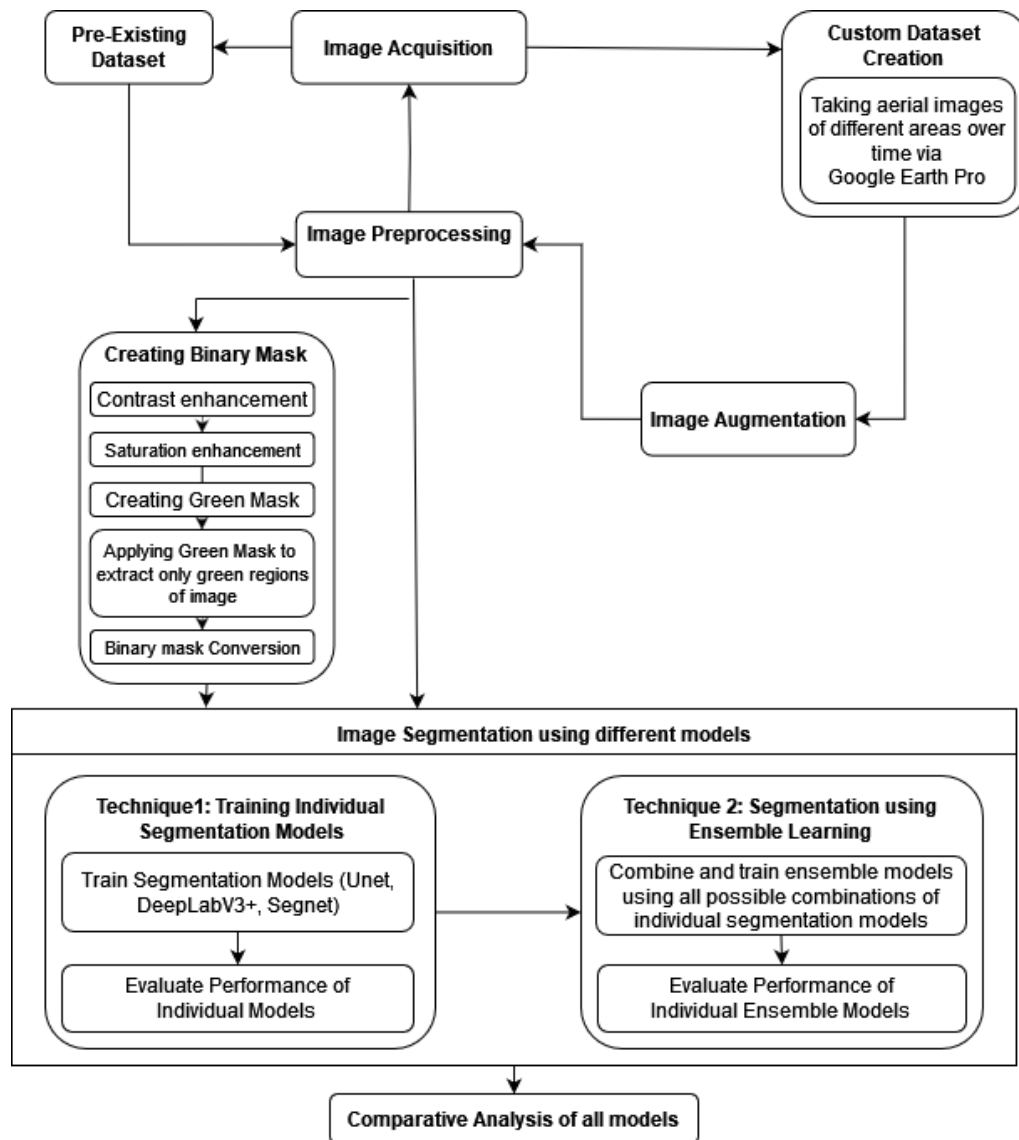


Figure 3.1: Methodology Framework for Detection

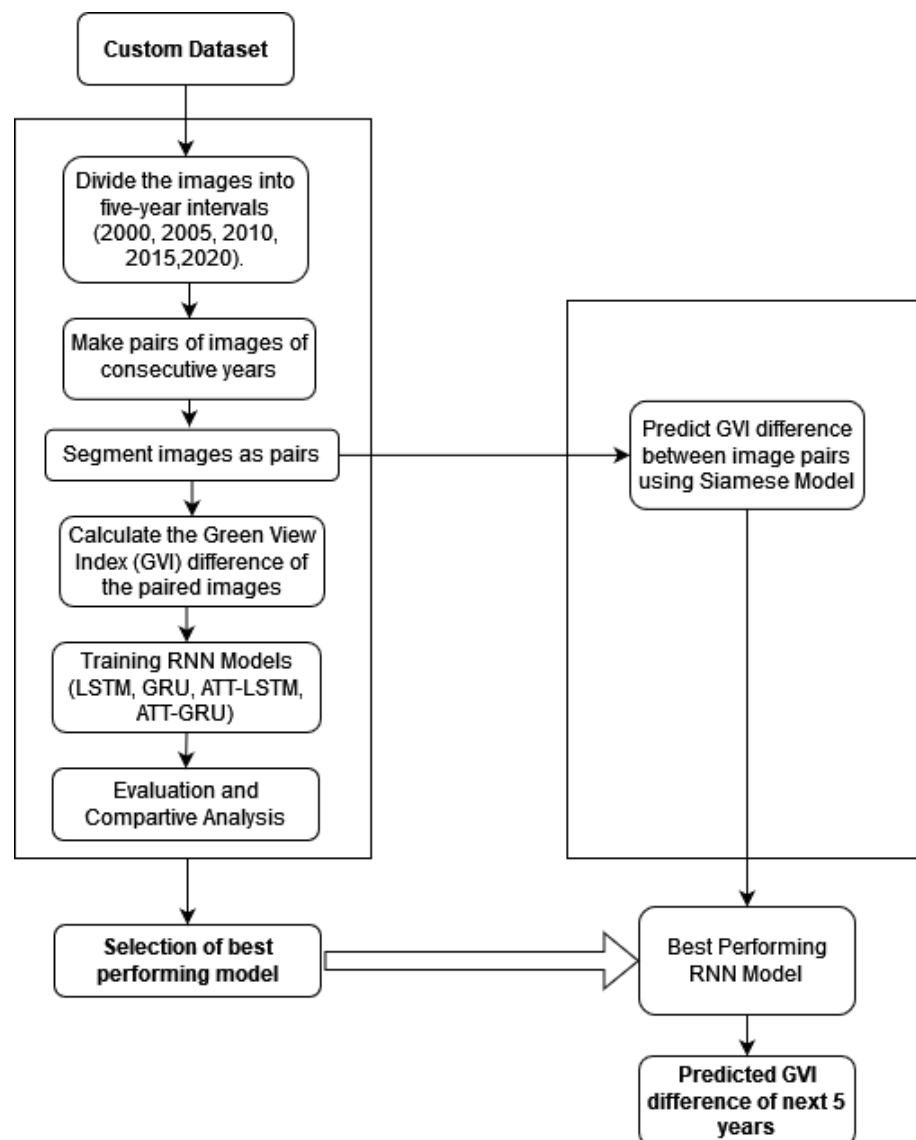


Figure 3.2: Methodology Framework for Prediction

potential disparities in dimensions are eliminated.

3.1.1 Kaggle Dataset

The "Forest Aerial Images for Segmentation" dataset from Kaggle was used. The dataset was obtained from the Land Cover Classification Track in DeepGlobe Challenge (?, ?). This dataset contains aerial images of locations with different coverage of land. It also has a variety of barren land and water bodies. It contained 5108 aerial forest images with dimensions of 256x256. Some example images from the Kaggle dataset are given below in Figure ???. It is visible from the images that the dataset contains a diverse range of land and green coverage and also includes water bodies of different depths and sizes.



Figure 3.3: Sample Images from Kaggle dataset

3.1.2 Custom Dataset

The custom data set was created from aerial images of different locations, captured over 20 years, from 2000 to 2020, using Google Earth Pro. After enabling historical imagery on Google Earth Pro, a slider is visible showing which years of images are available for the chosen location. Next, the slider is moved to the desired year and month (2000, 2005, 2010, 2015 and 2020). Notably, it was also attempted to maintain consistency in the month chosen for each year's set of images for the same location. Additionally, the view was selected to "Satellite" to get aerial imagery. Images were collected from countries like Canada (Vancouver), Australia (New South Wales, Victoria) and Brazil (Sao Paula) as images for these countries were available for as old as 2000. The custom data set is

subdivided into three categories, namely forest locations, residential locations, and parks.

Sample images from the custom dataset are shown in Figure ??



Figure 3.4: Sample Images from Categories of Custom Dataset

3.1.2.1 Image Augmentation

Several augmentation techniques that are specific to the images in the custom dataset are used to the data set to keep enhancing it. These methods include a wide range of geometric manipulations, including rotation, shifting, and flipping. For example, an image has been rotated 90 degrees clockwise, counter-clockwise, shifted in different positions, a few were flipped, etc. By using these various augmentation combination, the main motive is to increase the volume and diversity of the training data by applying various changes to the images. This augmentation method improves the segmentation models by emphasizing a deeper understanding of the various ways that green spaces can appear. As a result, the models have more stability, which makes it easier to identify and distinguish green locations in the images. A breakdown of the custom dataset is shown below in Table ??

Table 3.1: Breakdown of the Custom Dataset

SL. No	Category Name	Original Images	After Augmentation
1	Forest	155	510
2	Park	234	563
3	Residence	281	632
Grand total			1,705

3.1.3 Binary Mask Generation

To create a binary mask highlighting green vegetation in an image, several steps were undertaken. Initially, the image's contrast is enhanced to improve visual clarity, followed by saturation enhancement to amplify color information. Subsequently, a green mask is generated to isolate spaces with green hues. This mask is then applied to the original image, extracting only the green regions while removing non-green elements. Finally, the extracted green regions are converted into a binary mask, representing vegetation as white and non-vegetation as black pixels. This simplified process enables the isolation and identification of green vegetation within the image, facilitating further analysis or applications such as vegetation mapping and environmental monitoring.

3.1.4 Splitting into Train and Test Data

Splitting into train and test sets ensures that the models are trained on sufficient data while also allowing their performance to be assessed on unseen data. Both datasets is separately split into train and test sets with a ratio of 80% and 20% respectively. The test dataset with 20% of the images serves as a holdout set to assess the model's performance on unseen data.

3.2 Green Space Detection

In order to predict the green space of a location, it is first required to segment the green space from the images. Several segmentation models were explored which have been discussed further and as shown earlier in Figure ??, two techniques were used. This section describes the approaches applied to detect green space from satellite images.

3.2.1 Model Selection for Detection

This section discusses the segmentation models explored to segment green space from satellite images.

3.2.1.1 U-Net

Figure ?? portrays the architecture of U-Net used in this thesis work. The architecture of this model consists of an Input Layer, Contracting Path (encoder), Bottleneck, Expansive Path (decoder) and an Output Layer. In the contracting path, Convolutional blocks were used with increasing filter sizes and depths. Each convolutional block consisted of two 3x3 convolutional layers with ReLU activation and kernel initialization. MaxPooling2D was applied after the first convolutional layer in each block. Dropout layers were added after the first convolutional layer in each block for regularization. Additionally, the number of filters were doubled with each contracting block. In the bottleneck layer, a 3x3 convolutional block was used with dropout. In the Expansive Path, transposed convolutional layers with decreasing filter sizes and depths were used. Each convolutional block was mapped with the corresponding contracting path. Similar to the contracting path, convolutional layers with ReLU activation and kernel initialization were used along with dropout layers. The output layer contained a sigmoid activation function for binary image segmentation. Adam optimizer was used and Binary cross-entropy loss was used for binary image segmentation.

3.2.1.2 SegNet

SegNet is a convolutional neural network (CNN) architecture designed for semantic segmentation tasks. It consists of an Input Layer, Encoding Layers, Fully Connected (Dense) Layers, Decoding Layer, and an Output Layer. The encoding layer contained several

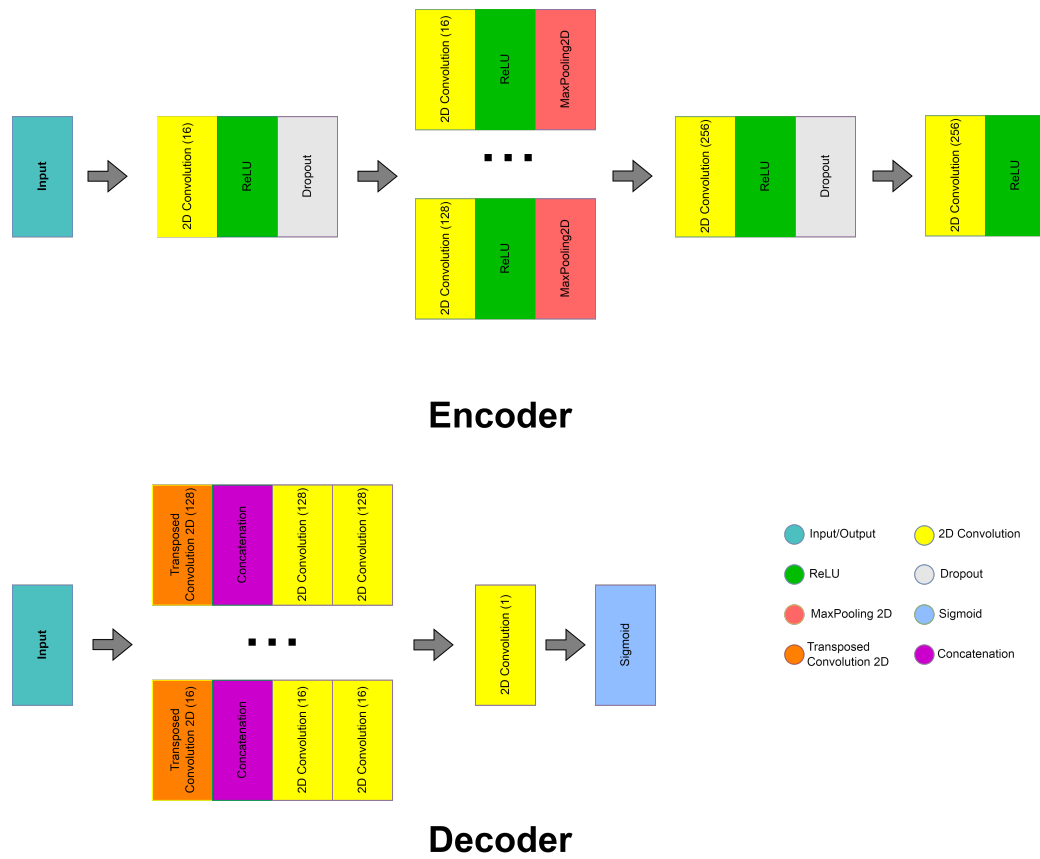


Figure 3.5: U-Net Model Architecture

convolutional layers with batch normalization and ReLU activation. Additionally, Max-Pooling2D layers were used for downsampling and the number of filters increased deeper into the network. After the encoding layers, two fully connected (dense) layers were added with ReLU activation (1024 units each). In contrast to encoding, the decoding layer consisted of upsampling, and layers were added to perform deconvolution, batch normalization, and ReLU activation was also applied. Additionally, the number of filters decreases in the decoding layers. Finally, the output layer contains one filter and a sigmoid activation function. In this architecture, adam optimizer was used with an initial learning rate of 0.001. The specific architecture used in this thesis work has been displayed in Figure ??.

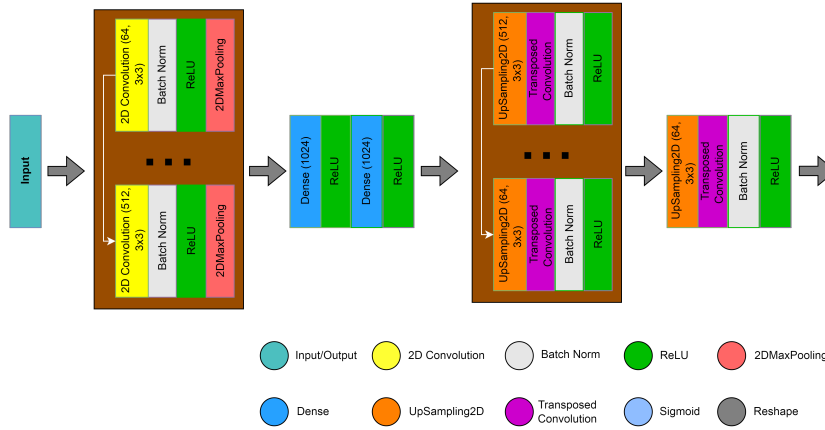


Figure 3.6: SegNet Model Architecture

3.2.1.3 DeepLabV3Plus

As shown below in Figure ??, DLV3Plus utilizes a ResNet50 backbone, a well-established deep learning architecture, to extract features from input images. Additionally, an Atrous Spatial Pyramid Pooling (ASPP) module is employed to capture contextual information at multiple scales, aiding in understanding objects of different sizes within an image. In the research, the model has been optimized using the Adam optimizer with a learning rate of

1×10^{-3} and binary cross-entropy loss, a common choice for binary classification tasks. Accuracy is monitored as a metric during training to evaluate the model's performance. This configuration ensures that the DLV3Plus model is skilled at accurately segmenting objects in images, contributing significantly to the research goals.

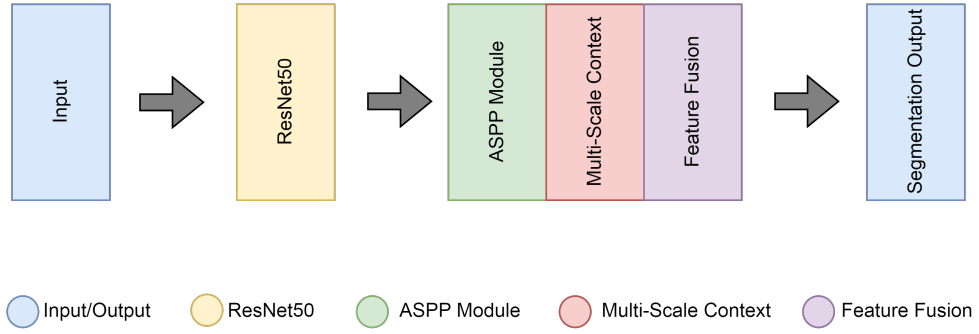


Figure 3.7: DeepLabV3plus Model Architecture

3.2.2 Technique 1: Training Different Segmentation Models

Both the pre-processed data set is fed into three different segmentation models: SegNet, UNet, and DeepLabV3+, along with the matching binary masks. The models were chosen for their established success in image segmentation tasks. SegNet efficiently handles high-resolution images, UNet demonstrates adaptability across diverse data types, and DeepLabV3+ has a powerful architecture designed for semantic segmentation. This ensures a thorough analysis of green space distribution in designated locations, enhancing the research's reliability. A thorough analysis of the distribution and composition of green spaces in the designated locations is made possible by the meticulous training of each model to precisely identify and segment green locations within the aerial images.

The models were trained using both the pre-existing data set and the custom data set separately to prove the robustness of the models.

3.2.3 Technique 2: Ensemble Learning-Based Approach

With this approach, multiple models are effectively combined to produce a single segmentation system. With the help of UNet, Segnet, and DeepLabV3 four ensemble models were created. The combinations of these four models are UNet-Segnet-DeepLabV3, UNet-Segnet, and UNet-DeepLabV3, and Segnet-DeepLabV3 respectively which is shown in Figure ???. The identical method used for training in technique 1 was also used for ensemble learning.

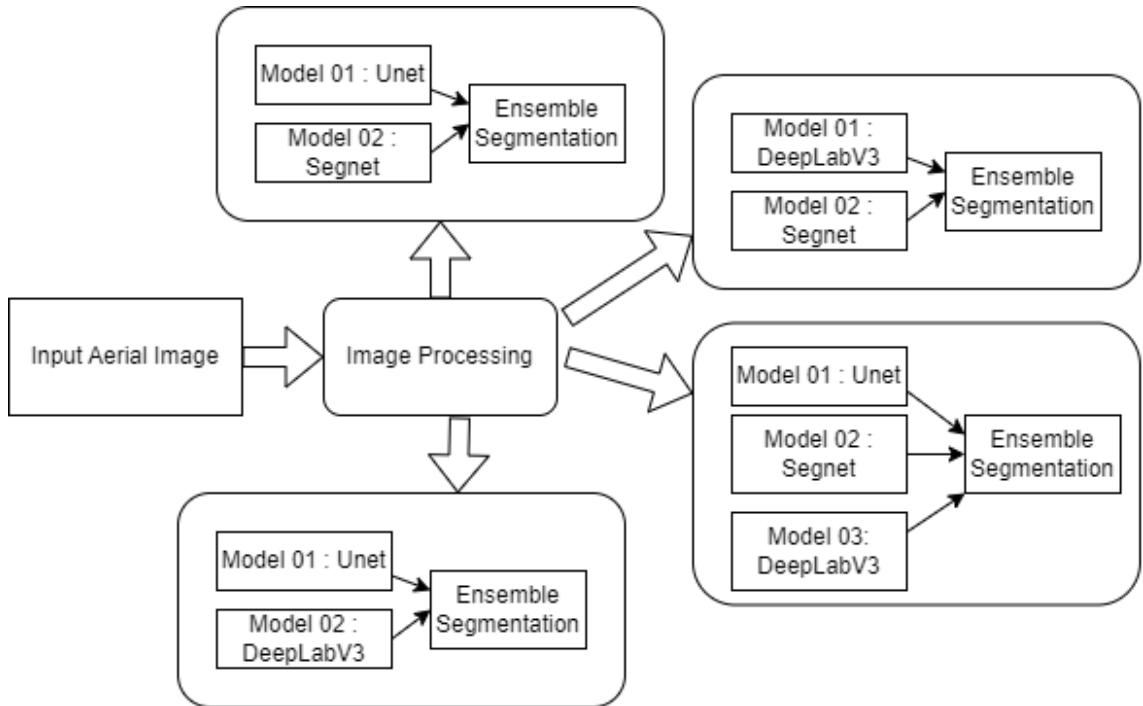


Figure 3.8: Framework for Ensemble Learning

3.3 Green Space Prediction

Green View Index (GVI) is a metric used in many research to measure the quantity of green space in a given location. For this reason, it has been chosen as the medium to measure the future amount of green space. In this research, binary segmented images have been used where 0 represents green space and 1 represents non-green space. Hence, based on the work by ? (?) and ? (?), the formula used by calculating GVI in this research is

$$\text{Green View Index (GVI)} = \frac{\text{Total number of white pixels in image}}{\text{Total pixels in image}} \quad (3.1)$$

However, rather than just determining the amount of green space, the focus of this research is to identify changes in green space. Therefore, throughout this research, the GVI difference between 5-year intervals has been employed as the data. The approach used for green space detection is covered in this section.

3.3.1 Image Pairing

The custom dataset contains images of the same location for a time range of 2000 to 2020, with an interval of five years. Firstly, the images were separated into designated years. Next, the images are made into pairs of consecutive images such as Pair P_i $\{Image_j, Image_k\}$ where j is recorded at an earlier timestamp than k . Therefore, for each location, there are four pairs of images:

- $Pair_1: \{Image_{2000}, Image_{2005}\}$
- $Pair_2: \{Image_{2005}, Image_{2010}\}$
- $Pair_3: \{Image_{2010}, Image_{2015}\}$

- $Pair_4: \{Image_{2015}, Image_{2020}\}$

This range of years was selected as it is not possible to calculate the ground truth value of years after 2024. In addition, images prior to 2000 are not easily accessible. Therefore, throughout this experiment, 2015 has been considered the present year and 2020 will be the future year following a five-year gap. The Green View Index (GVI) of each image is calculated and then the difference in the GVI between the paired images is calculated as

$$\Delta GVI = \text{GVI value of Image}_k - \text{GVI value of Image}_j \quad (3.2)$$

That means, if the GVI difference comes negative, there has been a decrease in green space and a positive GVI difference indicates an increase in green space over the next 5 years.

Finally, after pairing the images and computing their corresponding GVI differences, a redesigned dataset is produced which contains information about 341 locations. There are four data pairings for every place. Table ?? provides an illustration of how the data are organized.

Table 3.2: Redesigned Custom Dataset with Image Pairings

Pair	Image _j	Image _k	ΔGVI_{Pair_i}
P_1			30.02
P_2			-39.26
P_3			7.19
P_4			-8.73

3.3.2 Model Selection for Prediction

Recurrent Neural Network (RNN) models are capable of remembering information learned from prior input while generating outputs. Therefore, they are instrumental in sequential or time series data where each data represents a point in time. RNN models have two types of architecture - Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU). Moreover, adding attention mechanisms to the architectures has proven to show better results (?, ?, ?). Therefore, initially, four models have been used to predict the GVI difference over the next 5 years. The selected models have been elaborated further. The results of the models were compared and finally, the best-performing model has been selected for the proposed architecture. In this research, the GVI difference between the years 2000 to 2015 has been utilized to predict the GVI difference in the next 5 years (2015 to 2020). From the custom dataset, the GVI difference values for $Pair_1$, $Pair_2$ and $Pair_3$ are used as the feature variables and the GVI difference of $Pair_4$ has been used as the target variable. Each pair represents a different point in time. In total, the dataset has information of 341 locations. The RNN models have been then trained on this data.

3.3.2.1 Long Short-Term Memory (LSTM)

The architecture used in this research consists of two LSTM layers with 50 units each, followed by a dense layer with a single output neuron for GVI difference prediction. The model was compiled with separate loss functions for mean squared error (MSE) and mean absolute error (MAE), both optimized using the Adam optimizer.

3.3.2.2 Attention Long Short-Term Memory (ATT-LSTM)

The same LSTM model architecture was applied with an additional custom attention layer to add an attention mechanism. The attention mechanism was employed to give different

weights to different parts of the input sequence, emphasizing important information during the learning process. The attention mechanism utilizes trainable weights to compute attention scores for each time step in the LSTM output sequence. The attention scores are applied to the LSTM output sequence, giving higher importance to certain time steps.

3.3.2.3 Gated Recurrent Unit (GRU)

Similar to LSTM, the input layer of the Gated Recurrent Unit (GRU) model starts with 50 units and a "return sequences" parameter returns the full sequence of outputs for each input sequence. The second layer also consists of 50 units and a dense layer with a single output neuron was added for the GVI difference prediction.

3.3.2.4 Attention Gated Recurrent Unit (ATT-GRU)

The exact architecture of the GRU model was used in addition to a custom attention layer. This layer acts as an attention mechanism. In this layer, parameters (weights and biases) are learned during training. The attention mechanism computes attention scores for each time step, focusing relevant information in the input sequence.

3.3.3 Siamese Neural Network Model

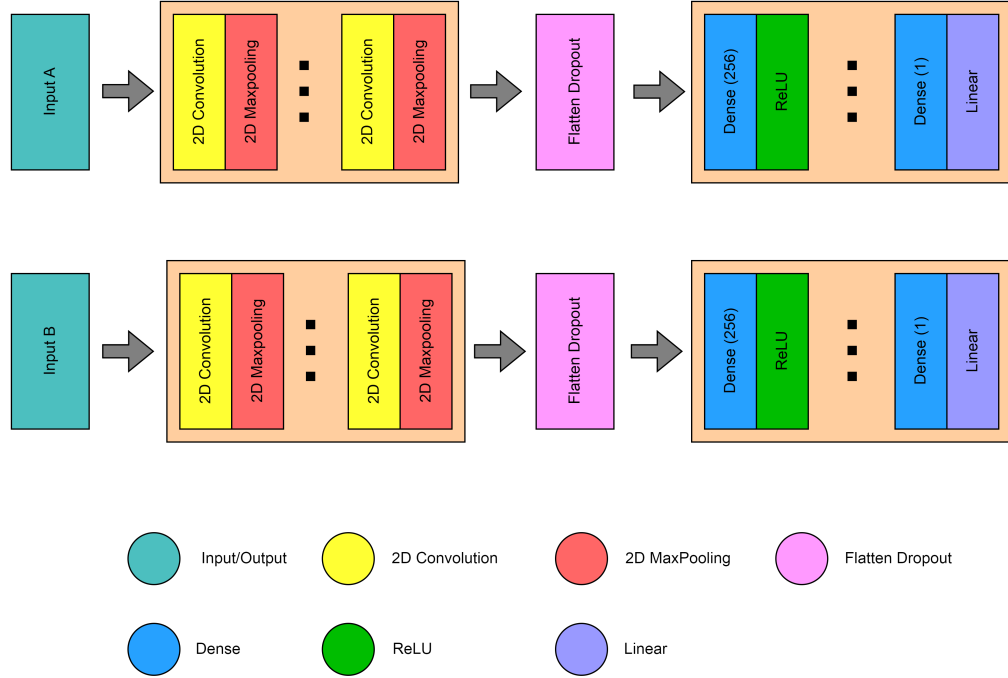


Figure 3.9: Siamese Neural Network Model Architecture

Siamese Neural Network is a reliable and robust architecture for detecting similarity between images and has been applied in various fields, from person identification (?, ?) to detecting change in land (?, ?). In this research, 4 individual siamese models were trained with segmented images of a specific pair $Pair_1, Pair_2, Pair_3,$

$Pair_4$. The segmented images used have been generated earlier in the green space detection phase of this research. So, a pair for the segmented image was given as the feature variables and the target variable was the GVI difference between the two images.

As shown in Figure ??, the Siamese model architecture used in this research consisted of shared convolutional layers followed by separate branches for each input image. The branches had fully connected layers, and the final output was obtained by concatenating

the outputs of these branches and passing through a regression output layer. The shared convolutional layer was made up of three layers containing 32,64 and 123 filters respectively of kernel size (3,3) along with ReLu activation function and MaxPooling layer. The applied architecture also consisted of a dropout layer with a dropout rate of 0.5 for regularization. The separate branches for each input image contained two fully connected layers with 256 units and ReLU activation for each input branch.

3.4 Proposed Framework

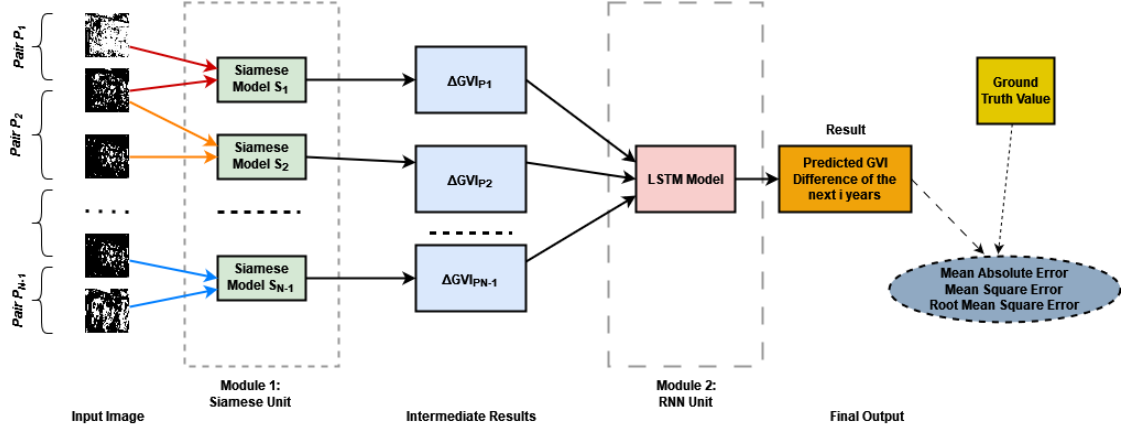


Figure 3.10: Proposed Framework for Predicting Change in GVI from Historical Images

The goal of this research is to predict the change in green space of a location in the next 5 years, based on past historical images of the said location. To do so, a two-module framework is proposed, as shown in Figure ???. Suppose there are N images of a selected location $\{y_T, y_{T+i}, \dots, y_{T+(n-1)i}\}$ where T is the earliest year, N is the number of images and i represents the interval considered (in this research, interval is considered as 5). These images are now made into pairs of non-overlapping consecutive years. So, from N images, there will be a total of $N-1$ Pairs.

Module 1, which is the Siamese unit, consist of multiple siamese model $S = \{S_1, S_2, \dots, S_n - 1\}$. Each pair P_i is sent to the i -th siamese model. Siamese model takes segmented

image of consecutive years as input and as output, gives a predicted GVI difference between the images. These are the intermediate result of the proposed framework.

In module 2, the RNN unit, the intermediate results achieved from module 1 are concatenated $\{\Delta\text{GVI}_{p_1}, \Delta\text{GVI}_{p_2}, \dots, \Delta\text{GVI}_{p_{N-1}}\}$ and sent to an LSTM model. Based on the result analysis in Chapter 4, the LSTM model is found to be the best-performing RNN model. LSTM is trained to learn long-term dependencies in sequential data. The model is given a range of GVI differences with an i -year gap (in this case, $i=5$), and based on this data, the final predicted GVI difference for the next i years is achieved.

CHAPTER 4

RESULT ANALYSIS

This chapter provides a detailed discussion of the findings and analysis

4.1 Evaluation Metrics

Evaluation metrics are measures used to assess the performance of machine learning models. The values of these metrics tell how well a model is performing. Evaluation metrics are essential for comparing different models or tuning their parameters.

4.1.1 Accuracy

Accuracy measures the proportion of correctly classified data out of the total number of data. It is the ratio of the number of correct predictions to the total number of predictions made by the model.

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Predictions}} \quad (4.1)$$

4.1.2 Precision

Precision is a metric that measures the number of true positive predictions out of all positive predictions made by the model. It is the ratio of true positives to the sum of true positives and false positives.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (4.2)$$

4.1.3 Dice Coefficient

The Dice coefficient is a metric used to evaluate the similarity or overlap between two sets. In the context of image segmentation, it measures the agreement between the predicted segmentation mask and the ground truth mask. The Dice coefficient ranges from 0 to 1, with higher values indicating greater overlap between the predicted and ground truth masks.

$$\text{Dice Coefficient} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.3)$$

4.1.4 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) is a metric used to measure the average magnitude of errors between predicted and actual values in regression tasks. It is the average of the absolute differences between predicted and actual values. MAE provides a straightforward interpretation of prediction accuracy with lower values indicating better performance.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4.4)$$

where y_i represents the true values, $\hat{y}_i$ represents the predicted values, and n represents the total number of samples.

4.1.5 Mean Square Error (MSE)

Mean Square Error (MSE) is a metric used to measure the average squared differences between predicted and actual values in regression tasks. It is the average of the squared differences between predicted and actual values. MSE penalizes larger errors more heavily than MAE which makes it sensitive to outliers in the data.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (4.5)$$

where y_i represents the true values, $\hat{y}_i$ represents the predicted values, and n represents the total number of samples.

4.1.6 Root Mean Square Error (RMSE)

Root Mean Squared Error (RMSE) is a measure used to quantify the average difference between predicted values and actual values in a dataset. RMSE is a popular metric in fields like statistics and machine learning, helping to measure how close predictions are to the actual values in a simple and understandable way.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4.6)$$

where y_i represents the true values, $\hat{y}_i$ represents the predicted values, and n represents the total number of samples.

4.2 Result and Performance Analysis for Green Space Detection

Table 4.1: Performance Comparison of Various Segmentation Models on Pre-Existing and Custom Dataset

Category	Pre-existing Dataset			Custom Dataset			
	Accuracy	Precision	Dice Co-efficient	Accuracy	Precision	Dice Co-efficient	
UNet	0.92	0.83	0.69	0.89	0.87	0.73	
Segnet	0.88	0.86	0.77	0.84	0.73	0.60	
DeepLabV3+	0.78	0.78	0.65	0.87	0.76	0.71	

4.2.1 Comparison with Pre-existing and Custom dataset

Table ?? shows the results for the performance comparison of the three state-of-the-art models on the entire pre-existing and custom dataset. It is seen from the table that, for the pre-existing dataset, U-Net is giving the highest accuracy but Segnet has the highest precision and dice-coefficient. On the other hand, for the entire custom dataset, UNet gives the highest accuracy, precision and dice coefficient.

Table 4.2: Performance Analysis of Various Segmentation Models on Custom Dataset Categories

Category	Park			Forest			Residence		
Model	Accuracy	Precision	Dice Coef- ficient	Accuracy	Precision	Dice Coef- ficient	Accuracy	Precision	Dice Coef- ficient
UNet	0.87	0.85	0.72	0.88	0.82	0.77	0.88	0.86	0.71
Segnet	0.74	0.51	0.59	0.71	0.48	0.37	0.81	0.67	0.56
DeepLabV3+	0.83	0.77	0.58	0.73	0.43	0.42	0.86	0.72	0.67

The custom dataset has been further broken down and the results for each category are shown in Table ?. For all three categories, U-Net is seen to give the highest accuracy, precision and dice coefficient. Another noteworthy observation is that for the Forest dataset, precision, and dice coefficient are significantly low for Segnet and DeepLabV3+ models as the forest dataset contains a much lower number of images among all the categories.

Although U-Net exhibited higher accuracy, Segnet is superior in both precision and dice-coefficient, portraying its ability to precisely identify as well as outline the area of interest through better localization. Thus, from the table and figures, it is evident that for the pre-existing dataset, Segnet outperforms the other two models while for the custom dataset, U-Net performs the best.

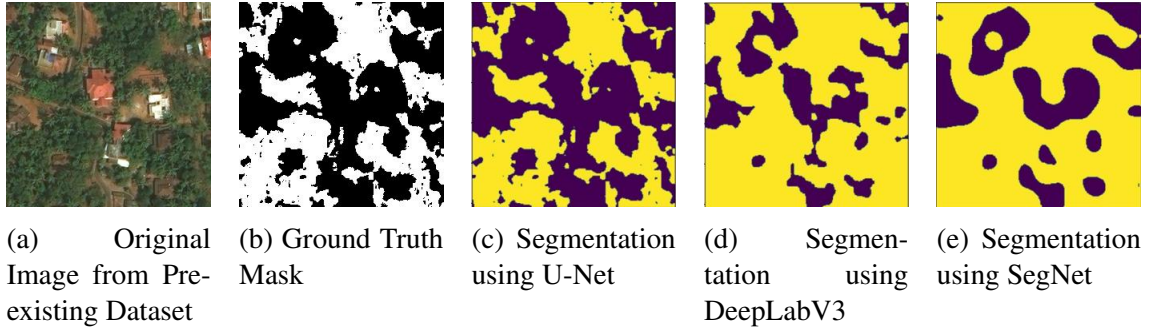


Figure 4.1: Outputs from Different Models for Pre-existing Dataset

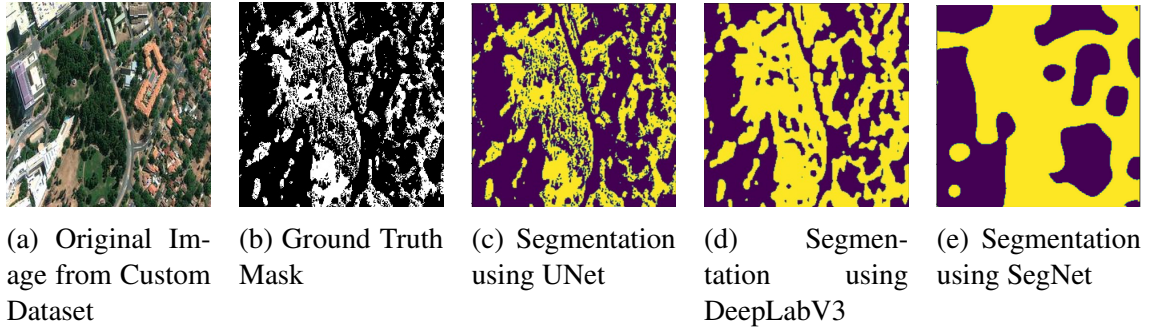


Figure 4.2: Outputs from Different Models for Custom Dataset

4.2.2 Comparative Analysis of Ensemble Learning Techniques

Table ?? shows the Accuracy, Precision, and Dice-Coefficient for all possible combinations of the deep learning models, that is ensemble learning. In the Preexisting dataset section of table ??, all the combination demonstrates the varying level of accuracy, precision, and dice coefficient for the collected data set with combination 4 (UNet-Segnet-DeepLabV3+) having the highest accuracy (0.84) and precision (0.96) among all the combinations. However, combination 3 (DeepLabV3+-Segnet) shows the highest dice-coefficient of 0.91 which is slightly more than the dice-coefficient of combination 4 (UNet-Segnet-DeepLabV3+). It is also evident that the performance of combination 2 (UNet-DeepLabV3+) has a significant discrepancy from the other combinations.

The custom dataset section of table ?? exhibits that combination 4 (UNet-Segnet-DeepLabV3+) has the highest accuracy (0.94) along with the highest precision (0.96) and dice coefficient (0.91). Additionally, it is observed that the performance

Table 4.3: Performance Analysis of Ensemble Learning on Pre-Existing and Custom Datasets

Combination	Pre-existing Dataset			Custom Dataset		
	Accuracy	Precision	Dice Co-efficient	Accuracy	Precision	Dice Co-efficient
UNet,Segnet	0.83	0.87	0.88	0.80	0.72	0.68
UNet, DeepLabV3+	0.25	0.35	0.50	0.83	0.88	0.57
DeepLabV3+, Segnet	0.65	0.93	0.91	0.81	0.95	0.77
UNet, Segnet and DeepLabV3+	0.84	0.96	0.82	0.94	0.96	0.91

of dice-coefficient of the models for the custom dataset is notably below standard compared to the pre-existing dataset.

From Tables ?? and ??, it is visible that SegNet excels in accuracy for the pre-existing dataset, while a combination of SegNet, UNet, and DeepLabV3+ performs best for the custom dataset. In terms of precision, the combination of SegNet, UNet, and DeepLabV3+ outperforms others on both datasets. For dice-coefficient, the combination of UNet and SegNet is best for the pre-existing dataset, and SegNet, UNet, and DeepLabV3+ excel for the custom dataset. In conclusion, the ensemble learning technique executes better than the deep learning techniques and the combination of SegNet, UNet and DeepLabV3 on custom dataset gives the most desirable result. Therefore it is proposed as the best model for segmenting green space from aerial images.

4.2.3 K-fold Cross-Validation

4.3 Result and Performance Analysis of Green Space Prediction

4.3.1 Comparative Analysis of RNN Models

Table 4.4: 5-fold Cross Validation Results

Test Set	Accuracy	Precision	Dice Coefficient
1	0.95	0.95	0.91
2	0.94	0.95	0.91
3	0.95	0.96	0.92
4	0.94	0.96	0.92
5	0.94	0.96	0.92
Average	0.94	0.96	0.92

Table 4.5: Test Metrics for Different RNN Models

Model	Test MAE	Test MSE	Test RMSE
GRU	0.7741	0.9600	0.9798
LSTM	0.5764	0.6216	0.7884
ATT-LSTM	1.600	5.0211	2.2408
ATT-GRU	1.888	5.586	2.3635

Table ?? portrays the evaluation results of the four RNN models performed. With a Test MAE of 0.7741, MSE of 0.9600 and an RMSE value of 0.9798, the GRU model showed a moderate performance and provided reasonable predictions. However, there is room for improvement in terms of minimizing errors. On the contrary, both Attention-based LSTM and Attention-based GRU exhibit higher error metrics compared to their basic architectures. Therefore, the application of attention mechanisms to the RNN architectures may not be the best choice for the required task. It is visible that the LSTM model outperforms the other models, achieving the lowest values across all three evaluation metrics. Thus, LSTM is proposed as the most fitting RNN model for predicting change in GVI for a selected location.

4.4 Experimental Evaluation

To evaluate the effectiveness of the proposed framework, historical images from 2000 to 2015 were used to forecast the same location's change in green space over the next five years. Thus, the GVI change from 2015 to 2020 was predicted using segmented photos of the same locations from 2000, 2005, 2010, and 2015. This range of years was selected as it is not possible to calculate the ground truth value of years after 2024. In addition, images prior to 2000 are not easily accessible. Therefore, in this experiment, 2015 has been considered as the current year and 2020 will be the future year following a five-year gap. The procedure followed and the results achieved from the experiment are presented in this section.

4.4.1 Experimental Procedure

The custom dataset was split into 80% train data and 20% test data. As the dataset contains segmented images from 4 different years with a five-year gap, the number of pairs created was 3, which are as follows:

- $Pair_1: \{Image_{2000}, Image_{2005}\}$
- $Pair_2: \{Image_{2005}, Image_{2010}\}$
- $Pair_3: \{Image_{2010}, Image_{2015}\}$

Each pair of test data was sent to a separate siamese model $\{S_1, S_2, S_3\}$, which was previously trained using segmented pictures from the chosen years. This is module 1 of the proposed framework. The exact siamese models trained earlier for $Pair_1$, $Pair_2$ and $Pair_3$ in Chapter 3 have been used in this experiment. From the Siamese models, the GVI difference value between the paired images $\{\Delta GVI_{P_1}, \Delta GVI_{P_2}, \Delta GVI_{P_3}\}$ were obtained were then fed to the LSTM model. This is the module 2 of the proposed framework.

Finally, the LSTM model gives a value as output which is the GVI difference between the latest year (2015) and the next 5 years (2020). The predicted GVI differences were then compared with ground truth values and metrics such as MAE, MSE, and RMSE were calculated.

4.4.2 Experimental Result

After experimenting as described in 4.4.1, an MAE of 0.2707, MSE of 0.1492, and an RMSE of 0.3862 was achieved. These results imply that the proposed framework, utilizing a combination of multiple Siamese neural networks and an LSTM model, performs reasonably well in forecasting the changes in green space. The low error metrics demonstrate the acquired precision and accuracy and highlight the model's capability to offer predictions about future changes in the GVI value of a given location.

CHAPTER 5

ENGINEERING CONSIDERATIONS, CHALLENGES AND REMEDIES

This chapter discusses the ethical, and environmental considerations and social constraints taken into consideration throughout this research work. The engineering activities involved in this thesis have also been elaborated.

5.1 Design Constraints

5.1.1 Ethical Considerations

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5.1.2 Sustainability in Environmental and Societal Context

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5.2 Complex Engineering Problem

5.2.1 Complex Problem Solving

Complex Engineering Problems addressed by this thesis have been justified in Table ??.

Table 5.1: Complex Engineering Problem (CEP)

Name of Complex Engineering Attribute	Justification
P1 Depth of knowledge required	For this thesis, in-depth knowledge of Artificial Intelligence, Deep Learning, Machine Learning and Digital Image Processing were required.
P3 Depth of analysis required	Several segmentation models and models suitable for time-series data were explored to propose a suitable model for both green space detection and prediction.
P4 Familiarity of issues	It was required to gather knowledge of different techniques and measures on how to utilise satellite imagery to measure green space, such as remote sensing and GVI calculation.
P7 Interdependence	This thesis work was done in two phases - green space detection and prediction. The prediction phase required segmented images that were achieved from the first phase. Additionally, the first and foremost task of this thesis was data acquisition, without which it is not possible to conduct the two phases.

5.2.2 Engineering Activities

Complex Engineering Activities addressed by this thesis have been justified in Table ??.

Table 5.2: Complex Engineering Activities (CEA)

Name of Complex Engineering Activity	Justification
A1 Range of Resources	Google Earth Pro and Kaggle were the resources for the datasets, Google Drive was utilized to store all data and necessary codes and Google Collaboartary environment was used to facilitate machine learning models.
A3 Innovation	After an extensive literature review, it was found that no work has been done that uses ensemble learning techniques to segment green space from images. Moreover, no work was found that used a time-series data analysis to predict change in green space. This thesis work is the first research that segments green spaces from images using an ensemble learning approach and then predicts green space.
A4 Consequences for Society and the Environment	Environment specialists, policymakers can use this proposed system to take necessary actions to restore the greenery of concerned areas. Urban planners can also make plans based on this information, taking into account areas that are most at risk of deforestation.

CHAPTER 6

DISCUSSION AND CONCLUSION

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6.1 Thesis Outcome

The outcome of this research is to

6.2 Thesis Implication and Contribution

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6.3 Thesis Limitation

It is crucial to acknowledge certain limitations. The following are some of the shortcomings of this thesis:

- ..
- ..
- ..
- ..

6.4 Future Work

The future work of this research is to

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